

D. $\frac{2}{\ln 2} \times \log_2(x^{2x-1})$

5. Calculus, 2ADV C2 SM-Bank 1 MC

If $f(x) = e^{g(x^2)}$, where g is a differentiable function, then $f'(x)$ is equal to

- A. $2xe^{g(x^2)}$
- B. $2xg(x^2)e^{g(x^2)}$
- C. $2xg'(x^2)e^{g(x^2)}$
- D. $2xg'(2x)e^{g(x^2)}$

6. Calculus, 2ADV C2 2008 HSC 2aii

Differentiate with respect to x :

$x^2\log_e x$ (2 marks)

7. Calculus, 2ADV C2 SM-Bank 13

Differentiate $y = 2e^{-3x}$ with respect to x . (2 mark)

8. Calculus, 2ADV C2 SM-Bank 7

Let $f(x) = \frac{e^x}{(x^2 - 3)}$.

Find $f'(x)$. (2 marks)

9. Calculus, 2ADV C2 SM-Bank 8

Let $y = (x + 5)\log_e(x)$.

Find $\frac{dy}{dx}$ when $x = 5$. (2 marks)

10. Calculus, 2ADV C2 2019 MET1 1a

Let $y = \frac{2e^{2x} - 1}{e^x}$.

Find $\frac{dy}{dx}$. (2 marks)

11. Calculus, 2ADV C3 2005 HSC 2d

Find the equation of the tangent to $y = \log_e x$ at the point $(e, 1)$. (2 marks)

12. Calculus, 2ADV C2 2009 HSC 2aii

Differentiate with respect to x .

$(e^x + 1)^2$. (2 marks)

13. Calculus, 2ADV C2 2011 HSC 1d

Differentiate $\ln(5x + 2)$ with respect to x . (2 marks)

14. Calculus, 2ADV C2 2012 HSC 11d

Differentiate $(3 + e^{2x})^5$. (2 marks)

15. Calculus, 2ADV C2 2012 HSC 12ai

Differentiate with respect to x

$(x - 1)\log_e x$ (2 marks)

16. Calculus, 2ADV C2 2013 HSC 11d

Differentiate x^2e^x (2 marks)

17. Calculus, 2ADV C2 2007 HSC 2ai

Differentiate with respect to x :

$\frac{2x}{e^x + 1}$ (2 marks)

18. Calculus, 2ADV C2 2015 HSC 11e

Differentiate $(e^x + x)^5$. (2 marks)

19. Calculus, 2ADV C2 2015 HSC 11f

Differentiate $y = (x + 4)\ln x$. (2 marks)

20. Calculus, 2ADV C2 2017 HSC 11d

Differentiate $x^3\ln x$. (2 marks)

21. Calculus, 2ADV C2 2018 HSC 11g

Differentiate $\frac{e^x}{x + 1}$. (2 marks)

22. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

$e^{\tan(2x)}$ (2 marks)

23. Calculus, 2ADV C2 EQ-Bank 1

Differentiate $\log_2 x^2$ with respect to x . (2 marks)

24. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

$$10^{5x^2-3x}. \quad (2 \text{ marks})$$

25. Calculus, 2ADV C2 EQ-Bank 3

Differentiate $3x 6^x$. (2 marks)

26. Calculus, 2ADV C2 2023 MET1 1a

Let $y = \frac{x^2 - x}{e^x}$.

Find and simplify $\frac{dy}{dx}$. (2 marks)

27. Calculus, 2ADV C2 SM-Bank 14

Differentiate π^{2x} . (2 marks)

28. Calculus, 2ADV C2 SM-Bank 4

Differentiate $5^{x^2} 5x$. (2 marks)

29. Calculus, 2ADV C2 SM-Bank 6

Differentiate with respect to x :

$$\log_e x^x. \quad (2 \text{ marks})$$

Worked Solutions

1. Calculus, 2ADV C2 2016 HSC 5 MC

$$y = \ln(\cos x)$$

$$\frac{dy}{dx} = \frac{-\sin x}{\cos x}$$

$$= -\tan x$$

$$\Rightarrow B$$

2. Calculus, 2ADV C2 2017 HSC 3 MC

$$y = e^{x^2}$$

$$\frac{dy}{dx} = 2x e^{x^2}$$

$$\Rightarrow C$$

3. Calculus, 2ADV C4 2018 HSC 5 MC

$$y = \sin(\ln x)$$

$$\frac{dy}{dx} = \cos(\ln x) \times \frac{d}{dx}(\ln x)$$

$$= \cos(\ln x) \times \frac{1}{x}$$

$$= \frac{\cos(\ln x)}{x}$$

$$\Rightarrow D$$

4. Calculus, 2ADV C2 EQ-Bank 4 MC

$$f(x) = \log_2(x^{2x})$$

$$= 2x \log_2 x$$

$$= \frac{2x \ln x}{\ln 2}$$

$$f'(x) = \frac{1}{\ln 2} \left(2x \cdot \frac{1}{x} + 2 \ln x \right)$$

$$= \frac{2}{\ln 2} + \frac{2 \ln x}{\ln 2}$$

$$= \frac{2}{\ln 2} + 2 \log_2 x$$

$$\Rightarrow B$$

5. Calculus, 2ADV C2 SM-Bank 1 MC

$$f(x) = e^{g(x^2)}$$

Using the chain rule (twice):

$$\begin{aligned} f'(x) &= \frac{d}{dx} [g(x^2)] \cdot e^{g(x^2)} \\ &= 2x \cdot g'(x^2) \cdot e^{g(x^2)} \end{aligned}$$

$$\Rightarrow C$$

6. Calculus, 2ADV C2 2008 HSC 2aii

$$y = x^2 \log_e x$$

$$\begin{aligned} \frac{dy}{dx} &= x^2 \cdot \frac{1}{x} + 2x \cdot \log_e x \\ &= x + 2x \log_e x \end{aligned}$$

7. Calculus, 2ADV C2 SM-Bank 13

$$y = 2e^{-3x}$$

$$\begin{aligned} \frac{dy}{dx} &= -3 \times 2e^{-3x} \\ &= -6e^{-3x} \end{aligned}$$

8. Calculus, 2ADV C2 SM-Bank 7

$$\text{Let } u = e^x \Rightarrow u' = e^x$$

$$v = (x^2 - 3) \Rightarrow v' = 2x$$

$$\begin{aligned} f'(x) &= \frac{e^x(x^2 - 3) - 2xe^x}{(x^2 - 3)^2} \\ &= \frac{e^x(x^2 - 2x - 3)}{(x^2 - 3)^2} \end{aligned}$$

9. Calculus, 2ADV C2 SM-Bank 8

$$\begin{aligned} \frac{dy}{dx} &= 1 \times \log_e x + (x + 5) \cdot \frac{1}{x} \\ &= \log_e x + \frac{x + 5}{x} \end{aligned}$$

$$\therefore \frac{dy}{dx} \bigg|_{x=5} = \log_e 5 + 2$$

10. Calculus, 2ADV C2 2019 MET1 1a

Method 1

$$y = 2e^x - e^{-x}$$

$$\frac{dy}{dx} = 2e^x + e^{-x}$$

Method 2

$$\begin{aligned} \frac{dy}{dx} &= \frac{4e^{2x} \cdot e^x - (2e^{2x} - 1)e^x}{(e^x)^2} \\ &= \frac{4e^{3x} - 2e^{3x} + e^x}{e^{2x}} \\ &= \frac{2e^{2x} + 1}{e^x} \end{aligned}$$

11. Calculus, 2ADV C3 2005 HSC 2d

$$y = \log_e x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

At $(e, 1)$,

$$m = \frac{1}{e}$$

Equation of tangent, $m = \frac{1}{e}$, through $(e, 1)$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{e}(x - e)$$

$$y - 1 = \frac{x}{e} - 1$$

$$y = \frac{x}{e}$$

12. Calculus, 2ADV C2 2009 HSC 2aii

$$y = (e^x + 1)^2$$

$$\begin{aligned} \frac{dy}{dx} &= 2(e^x + 1)^1 \times \frac{d}{dx}(e^x + 1) \\ &= 2e^x(e^x + 1) \end{aligned}$$

13. Calculus, 2ADV C2 2011 HSC 1d

$$y = \ln(5x + 2)$$
$$\frac{dy}{dx} = \frac{5}{5x + 2}$$

14. Calculus, 2ADV C2 2012 HSC 11d

$$y = (3 + e^{2x})^5$$
$$\frac{dy}{dx} = 5(3 + e^{2x})^4 \times \frac{d}{dx}(3 + e^{2x})$$
$$= 5(3 + e^{2x})^4 \times 2e^{2x}$$
$$= 10e^{2x}(3 + e^{2x})^4$$

15. Calculus, 2ADV C2 2012 HSC 12ai

$$y = (x - 1)\log_e x$$
$$\frac{dy}{dx} = 1(\log_e x) + (x - 1)\frac{1}{x}$$
$$= \log_e x + 1 - \frac{1}{x}$$

16. Calculus, 2ADV C2 2013 HSC 11d

Using the product rule

$$\text{Let } u = x^2, \quad u' = 2x$$

$$\text{Let } v = e^x, \quad v' = e^x$$

$$\frac{d(uv)}{dx} = u'v + v'u$$
$$= 2xe^x + x^2e^x$$
$$= xe^x(x + 2)$$

17. Calculus, 2ADV C2 2007 HSC 2ai

$$y = \frac{2x}{e^x + 1}$$
$$u = 2x \quad v = e^x + 1$$
$$u' = 2 \quad v' = e^x$$
$$\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$
$$= \frac{2(e^x + 1) - 2x(e^x)}{(e^x + 1)^2}$$
$$= \frac{2e^x + 2 - 2x \cdot e^x}{(e^x + 1)^2}$$
$$= \frac{2(e^x + 1 - xe^x)}{(e^x + 1)^2}$$

18. Calculus, 2ADV C2 2015 HSC 11e

$$y = (e^x + x)^5$$
$$\frac{dy}{dx} = 5(e^x + x)^4 \times \frac{d}{dx}(e^x + x)$$
$$= 5(e^x + x)^4 \times (e^x + 1)$$
$$= 5(e^x + 1)(e^x + x)^4$$

19. Calculus, 2ADV C2 2015 HSC 11f

$$y = (x + 4)\ln x$$

Using the product rule

$$\frac{dy}{dx} = \frac{d}{dx}(x + 4) \cdot \ln x + (x + 4)\frac{d}{dx}\ln x$$
$$= \ln x + (x + 4)\frac{1}{x}$$
$$= \ln x + \frac{4}{x} + 1$$

20. Calculus, 2ADV C2 2017 HSC 11d

$$y = x^3 \ln x$$

Using the product rule:

$$\frac{dy}{dx} = 3x^2 \cdot \ln x + x^3 \cdot \frac{1}{x}$$
$$= x^2(3\ln x + 1)$$

21. Calculus, 2ADV C2 2018 HSC 11g

$$y = \frac{e^x}{x+1}$$

Differentiate using quotient rule:

$$u = e^x \quad v = x+1$$

$$u' = e^x \quad v' = 1$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{e^x(x+1) - e^x \cdot 1}{(x+1)^2} \\ &= \frac{xe^x}{(x+1)^2} \end{aligned}$$

22. Calculus, 2ADV C2 EQ-Bank 2

$$y = e^{\tan(2x)}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \tan(2x) \times e^{\tan(2x)} \\ &= 2\sec^2(2x) \cdot e^{\tan(2x)} \end{aligned}$$

23. Calculus, 2ADV C2 EQ-Bank 1

$$y = \log_2 x^2$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{\ln x^2}{\ln 2} \right) \\ &= \frac{1}{\ln 2} \cdot \frac{d}{dx} (\ln x^2) \\ &= \frac{1}{\ln 2} \cdot \frac{2x}{x^2} \\ &= \frac{2}{x \ln 2} \end{aligned}$$

TIP: The new Advanced reference sheet can be used here!

24. Calculus, 2ADV C2 EQ-Bank 2

$$y = 10^{5x^2-3x}$$

$$\frac{dy}{dx} = \ln 10 (10x-3) \cdot 10^{5x^2-3x}$$

TIP: The new Advanced reference sheet can be used here!

25. Calculus, 2ADV C2 EQ-Bank 3

$$y = 3x \cdot 6^x$$

$$\begin{aligned} \frac{dy}{dx} &= 3 \cdot 6^x + \ln 6 \cdot 6^x \cdot 3x \\ &= 3 \cdot 6^x (1 + x \ln 6) \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 6^x .

26. Calculus, 2ADV C2 2023 MET1 1a

Using the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{e^x(2x-1) - (x^2-x)e^x}{(e^x)^2} \\ &= \frac{e^x(-x^2+3x-1)}{e^{2x}} \\ &= \frac{-x^2+3x-1}{e^x} \end{aligned}$$

27. Calculus, 2ADV C2 SM-Bank 14

$$y = \pi^{2x}$$

$$\begin{aligned} \frac{dy}{dx} &= \log_e(\pi) \cdot 2 \cdot \pi^{2x} \\ &= 2\log_e(\pi) \cdot \pi^{2x} \end{aligned}$$

COMMENT: π is a constant. See HSC exam reference sheet for differentiating $a^{f(x)}$.

28. Calculus, 2ADV C2 SM-Bank 4

$$y = 5^{x^2} \cdot 5x$$

$$\begin{aligned} \frac{dy}{dx} &= \ln 5 \cdot 2x \cdot 5^{x^2} \cdot 5x + 5^{x^2} \cdot 5 \\ &= 5^{x^2} (\ln 5 \cdot 10x^2 + 5) \\ &= 5^{x^2+1} (\ln 5 \cdot 2x^2 + 1) \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 5^x .

29. Calculus, 2ADV C2 SM-Bank 6

$$y = \log_e x^x$$

$$= x \log_e x$$

$$\begin{aligned} \frac{dy}{dx} &= x \cdot \frac{1}{x} + \log_e x \\ &= 1 + \log_e x \end{aligned}$$

